Structure, Not Texture: Train-Free, World-Knowledge-Grounded Interiors for Gaussian Splatting

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Abstract

When a 3D Gaussian-Splatting object is cut, broken, or sliced open, what does its interior look like? A splat is a hollow shell of surface primitives, so every existing answer *fabricates* the inside. PhysGaussian and PhysTalk deform only the surface and never expose an interior; FruitNinja hallucinates interior *texture* with a per-object diffusion optimization; GaussianFluent densifies interior gaussians with a diffusion prior in the loop. All of these priors are *learned or optimized*, blind to the identity of the specific object, and slow. We observe that a vision–language model already *knows* what is inside a watermelon, a tree log, or an onion, and that this knowledge is a *train-free, structural* prior. We present InsideOut, a framework in which a VLM agent recognizes an object, reasons about its internal layout, and *writes a short program* that instantiates that structure directly in the gaussians—radial seeds, concentric rings, nested layers—which a topology-changing operation then reveals. There is no training, no per-object optimization, and no diffusion in the loop: interiors are authored in seconds and stay consistent under arbitrary cut directions. Because it is instant, the method scales editable, physically-changeable interiors from single objects to whole scenes—a step toward explorable worlds whose objects can be opened, cut, and broken. We demonstrate structurally-correct interiors across [TODO: N] objects and [TODO: M] operations, head-to-head against FruitNinja (structure vs. texture, zero-optimization runtime), GaussianFluent (no diffusion-in-loop), and PhysTalk (interior vs. none). We are explicit about the failure mode we inherit: when an object's interior is genuinely unidentifiable, the agent should *abstain* rather than confabulate.

1 Introduction

(Arc: (1) worlds from media should be changeable, not just viewable; (2) changing an object means exposing its inside; (3) splats are hollow shells, so the inside is always invented; (4) prior inventions are learned/optimized/slow/category-blind; (5) the VLM already knows the inside — use it as a train-free structural prior; (6) contributions.)

[TODO: one motivating figure: same watermelon splat cut open — PhysTalk (deflates, no interior) | FruitNinja-style texture fill (smeared, wrong seed placement, cut-inconsistent) | ours (radial seeds placed by world knowledge, consistent under any cut).]

Contributions.

- We reframe an object's interior as recoverable from a VLM's *world knowledge*, used as a *train-free structural prior* — in place of the learned, optimized, or diffusion priors of prior work.
- An *agentic, train-free* pipeline: recognize → reason about internal layout → author a program over an interior/physics op-library → instantiate structured interior gaussians → reveal via a topology-changing operation. No per-object optimization, no diffusion in the loop.
- *Structure, not texture*: explicit 3D internal geometry (seeds, rings, layers) that is *consistent under any cut direction by construction*, unlike per-cross-section inpainting.

- A demonstration across objects and operations with direct head-to-heads against FruitNinja, GaussianFluent, and PhysTalk, and an honest treatment of *unidentifiable* interiors (abstention), connecting to identifiability in single-view 3D generation.

2 Related Work

(Position surgically — the gap is narrow; each neighbor is one axis away. State plainly what each does and does NOT do.)

Physics on Gaussian Splats. PhysGaussian, PhysSplat, and **PhysTalk** embed 3DGS in a physics simulator (MPM / Genesis) and deform the visible surface. PhysTalk is the closest on the *agentic, train-free* axis—an LLM writes physics code—but it never instantiates or reveals an interior; it transfers particle motion back to existing surface gaussians. [TODO: cite; confirm no topology-change / interior].

Interior generation for splats. FruitNinja (CVPR'25) first revealed a splat interior under slicing, via *per-object* progressive 2D-diffusion inpainting on cross-sections plus voxel smoothing. It is not agentic, its prior is category-blind, and its result is *texture* with no guarantee that seeds/rings sit where they should, nor that different cuts agree. **GaussianFluent** (ICLR'26) densifies interior gaussians for brittle *fracture*, but with a *diffusion prior in the loop*, not agentic and not identity-grounded. [TODO: cite both; this is the head-to-head axis.]

Inferring interior structure. TopoGaussian infers an object's *load-bearing* interior topology by optimizing a representation so a simulator explains observed motion—per-object optimization, not semantic ("what mechanically works", not "this is a watermelon"). [TODO: cite.]

Agentic program authoring for 3D. Articraft (2026) has an LLM agent write a program against a DSL to build articulated 3D *meshes*. It establishes "agent-writes-a-program-for-3D" as a paradigm—so that framing alone is not our novelty—but it does not touch captured splats, interiors, or topology-changing reveals. [TODO: cite; use to pre-empt "just Articraft for splats".]

Editable / language-driven splats. [TODO: REALM (MLLM-agent surface editing of 3DGS, no interior), VR-GS (tet-proxy interactive physics), etc.]

3 Method

3.1 Overview

From a single photograph, an off-the-shelf image-to-3DGS model (TripoSplat) reconstructs the object's exterior as a Gaussian *shell* with a hollow interior. Given the object and an instruction ("cut it open"), a vision–language model (VLM) authors an *interior schema* describing the object's internal structure. We instantiate that schema as a *gapless* volume of interior gaussians, then *fit* their appearance by differentiable optimization against a perceptual signal, evaluated over many cut directions. A cut, break, or slice then reveals a believable interior that is consistent from any direction. Nothing is trained: we compose frozen off-the-shelf models (TripoSplat, the VLM, and a frozen perceptual/CLIP prior) with procedural code.

3.2 World knowledge as a structural prior

The VLM reads the object and emits a compact JSON *interior schema*: an ordered list of `layers` by normalized depth-from-surface (each with an RGB colour), plus discrete `inclusions` (`seeds`, `core`, `pit`) with a count, size, colour, and depth region. For a watermelon it writes a pale rind over red flesh with a few hundred black seed inclusions; for a kiwi, a brown skin over green flesh with a pale central core and a ring of small black seeds. This schema is the *train-free prior*: it encodes what the VLM knows about the category and is correct by construction, not by inpainting luck. Because the VLM is a *vision* model, it can also inspect a rendered cut and revise its own schema (a write–render–look–fix loop).

3.3 Instantiating a gapless structured interior

Reconstructed shells are thin and *non-watertight*, so a flood-fill solid test leaks (on our objects it recovered near-zero enclosed voxels). We instead mark a voxel as interior when the surface encloses it along at least two of the three axes—leak-proof, and it fills the concavities that a strict three-axis test leaves as oblique-cut voids. Each interior voxel becomes one gaussian, jittered off the lattice with anisotropic scale and random orientation so the fill reads as native splats rather than a voxel grid. We colour by the schema—layers by depth, inclusions as discrete clusters (seed count scales with fill density so seeds stay visible)—and *never* by copying surface colour inward, the failure mode of texture fills.

3.4 Fitting the interior by differentiable optimization

The instantiated interior is structurally correct but flat. Since the Gaussian rasterizer is differentiable, we *fit* the interior gaussians (colour, scale, opacity; surface frozen) by gradient descent against a perceptual signal *averaged over many randomly sampled cut planes*. When a real cross-section photograph of the category exists we use a DINOv2 feature-matching loss to it; otherwise the loss is a CLIP image–text similarity to a prompt ("a cross-section of a kiwi…"), which needs no reference. A small anchor to the schema colours prevents drift. Optimising over *many* cut directions is what forces a coherent 3D interior—a structure that only looked right from one angle is punished by the others—so cut-anywhere realism is the *result* of the objective, not a hope. Crucially the prior guides a single 3D volume and never inpaints a 2D cut face, which keeps us distinct from per-cut diffusion methods.

3.5 Revealing the interior

Because the interior is authored in 3D before any cut, a cut/break/slice is a geometric operation on a fully-specified object: any plane exposes a valid, mutually-consistent cross-section, with no per-cut generation and no latency.

4 Experiments

(Framed as a CAPABILITY DEMONSTRATION, not a benchmark. NO ground-truth interior exists (FruitNinja says so) — we do NOT chase faithfulness-to-truth. Success = structure beats texture, train-free.)

4.1 Success criteria

- **Structural correctness** (primary): do placed elements match category-expected layout (seeds radial, rings concentric, layers nested)? Blind human + VLM-judge (randomized order, report both directions — VLM judges have position bias), plus a geometric check (discrete 3D seeds at plausible positions vs. smeared texture).
- **Cut-consistency**: same object, multiple cuts → interior stays 3D-consistent (no GT needed).
- **Efficiency (two regimes)**: the instant VLM-schema fill needs *no* optimization (seconds); the metric-fit interior is a short per-object optimization (a few minutes on one GPU) that buys realism—still orthogonal to, and lighter than, inpainting every cut with a diffusion model.

4.2 Dataset

We evaluate on captured objects reconstructed with TripoSplat, weighted to visibly-structured interiors. Our current set is *watermelon* (red flesh + seeds), *red onion* (concentric scales), *apple* (cream flesh + core), and *kiwi* (green flesh + pale star core + seed ring); every schema is authored by a real VLM from the whole-object photo. We are expanding toward ∼30–40 objects spanning radial seeds, concentric rings/layers and segmented interiors, plus deliberately *ambiguous* interiors (a sealed device, a random rock) for the abstention/failure analysis. There is no per-instance ground truth; category-reference cut photographs are used only for judging.

4.3 Head-to-heads

[TODO: vs FruitNinja (structure vs. texture + runtime); vs GaussianFluent (no diffusion-in-loop, fracture); vs PhysTalk (interior vs. none). Skipping any → reviewer reduces us to that method.]

4.4 Results

Across the four objects the method produces recognizable cross-sections from every cut direction: red flesh with seeds, onion rings, and a kiwi's star core and seed ring. The differentiable fit visibly sharpens the flat schema fill into realistic, tonally-varied material while preserving the authored structure. The reference-less CLIP signal is reliable on strongly-structured, strongly-coloured interiors but can introduce colour artifacts on *uniform* interiors (apple)—the honest hard case, where a diffusion (SDS) prior is the natural remedy. A live, in-browser demo lets one cut each object open and orbit the exposed interior. [TODO: quantitative human/VLM-judge table; runtime numbers; head-to-heads.]

5 Limitations

When an object's interior is genuinely *unidentifiable* from its exterior (a sealed device, an arbitrary rock), no prior—VLM or otherwise—can recover it; the agent should *abstain* rather than confabulate. (This is where the unidentifiability thesis of our prior work reappears — as an honest limitation, not the paper's thrust.) Our interiors are *plausible and structurally category-correct*, not faithful to the specific instance; and the VLM's knowledge has gaps for rare or novel objects.

Two limitations we observed in practice. First, reconstruction quality bounds us: a cluttered input photograph leaves background fragments that the matting step does not fully remove, which our fill then treats as spurious interior—the method wants clean, single-object images. Second, our depth-parameterized schema captures concentric and radial structure (rings, layers, cores, and scattered seeds) but not *angular* structure such as an orange's wedge segments or an apple's five-point seed star; an angular/segment schema primitive is natural future work.

6 Conclusion

[TODO: One instant, train-free agent authors structured interiors from world knowledge — the enabling piece for scaling changeable objects from one to a whole explorable world.]

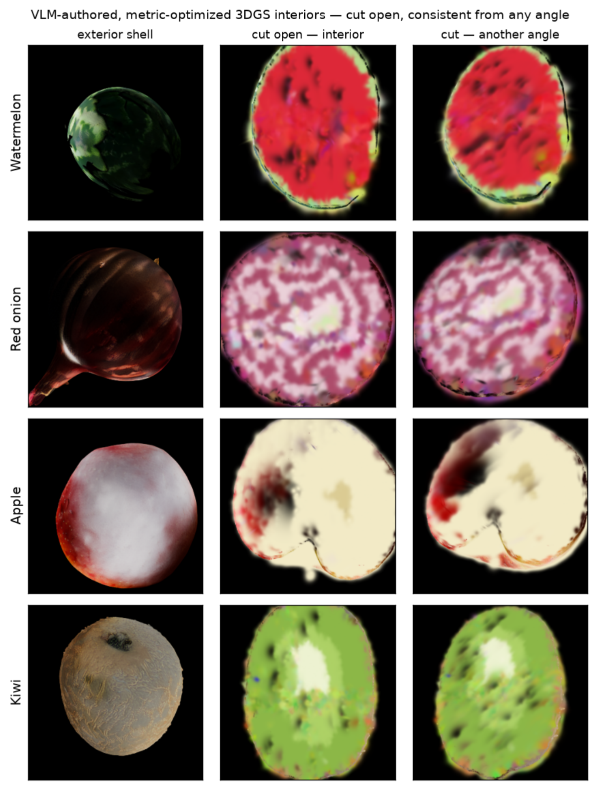

Figure 1: VLM-authored, metric-optimized interiors on four captured objects. Each row: the reconstructed exterior shell (left) and the object cut open from two directions (right), revealing the interior our method authors and fits—watermelon flesh and seeds, onion concentric rings, apple flesh and core, kiwi green flesh with a pale star core and a seed ring. The two cut views of each object agree: the interior is a single coherent 3D volume, so any cut is consistent (cut-anywhere). Apple is the honest hard case (uniform interior).